

Scalability & Future Plan

Date	June 26, 2026
Team ID	SWTID-2026-6095
Project Name	EduGenie
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No persistent database; the system uses a stateless design	Users lose quiz history and learning progress once a session ends	Medium
2	Core generation relies on the cloud-based Gemini AI service	Response speed and availability depend on internet connectivity and third-party API uptime	High
3	Single-instance deployment optimized for lightweight/local use (e.g. Mac M1)	Limited concurrent user capacity; not yet designed for large-scale simultaneous traffic	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user, local deployment; handles one session at a time	Deploy behind Uvicorn/Gunicorn workers with horizontal scaling and a load balancer to support many concurrent users
2	Data Storage	No persistent database; stateless in-memory session handling	Introduce SQLite for pilot use, migrating to PostgreSQL/MongoDB in production to store user profiles, quiz history, and learning progress
3	Performance	Direct, synchronous calls to Gemini AI per request	Add response caching (e.g. Redis) for repeated queries/explanations and use asynchronous request handling to reduce latency
4	Security	No authentication; open access to all endpoints	Implement JWT/OAuth2-based authentication, request rate limiting, and secure API key management for the Gemini AI integration

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Persistent user accounts & login (JWT-based auth) with a lightweight database for progress tracking	Q3 2026	Enables saved quiz history, personalized learning progress, and a returning-user experience
Phase 3	Multi-language support for explanations, quizzes, and summaries	Q4 2026	Broadens accessibility for non-English speaking learners
Phase 4	Mobile app (iOS/Android) and an educator analytics dashboard	Q1 2027	Increases reach for on-the-go learning and gives educators visibility into student progress